Gokulakrishna K

Senior Software Engineer – Backend | Java & Distributed Systems | Applied AI (RAG, MCP)
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Professional Summary

Senior Software Engineer with 5 years of experience building distributed, fault-tolerant backend microservices in Java and Spring Boot across healthcare and CRM products serving 250K+ users. Author of SynapseMCP, an open-source multi-tenant RAG + Model Context Protocol (MCP) platform, extending that systems rigor into applied AI infrastructure. Strong in system design, Domain-Driven Design, REST APIs, concurrency, and multi-tenant architecture.

Technical Skills

Languages: Java (Primary), Python, C++, SQL (PostgreSQL, MySQL).
Architecture & Concepts: Data Structures & Algorithms, REST APIs, Microservices, Distributed Systems, Concurrency, System Design.
AI Engineering: Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), LLM, Embeddings, Hybrid Retrieval.
Frameworks: Spring Boot, Spring AI, Spring Data JPA, JUnit, Struts.
Databases & Caching: PostgreSQL, MySQL, Redis.
Cloud & DevOps: AWS (EC2, EKS, SQS, S3, VPC, Lambda), Docker, Kubernetes, Jenkins, Maven, Gradle, Terraform, Git.
Observability & Monitoring: Amazon CloudWatch, Prometheus.

Experience

CitiusTech (Client: athenahealth) August 2025 – Present
Senior Software Engineer (Backend) *Chennai, India*

- Redesigned core healthcare domain models for Medical Record Release (MRR) document reconciliation using **Domain-Driven Design**, resolving scalability and data-consistency gaps across the independently deployed message-generation and document-update microservices.
- Built hexagonal-architecture integrations with IAM authentication and idempotency/retry semantics for the external Medical Document Platform (MDP) APIs, sustaining fault-tolerant exchange of **2K+ documents monthly** in production.
- Developed REST APIs transforming **FHIR** Audit Events into vendor messages, establishing the record-level regulatory compliance required across the multi-system reconciliation pipeline.
- Implemented a domain-level validation framework separating terminal from transient failures, blocking invalid data from propagating downstream and cutting retry-related incidents from **~2/week to ~1/month** by making automated retries safe by default.
- Introduced MDC-based structured logging with distributed tracing across services, cutting cross-service incident diagnosis from manual log correlation across systems to following a single trace ID.

Zoho (Zoho CRM) August 2021 – August 2025
Member Technical Staff (Backend Engineer) *Chennai, India*

- Owned the Module Builder – Fields component serving **250K+ users**, evolving the field-metadata architecture behind custom modules, layouts, sections, and fields through four years of stable releases.
- Built dynamic field-limit expansion and on-demand field-type conversion tooling, eliminating a hard product limit and the recurring per-customer escalations it caused.
- Implemented sandbox support for the full field-configuration lifecycle (create, update, delete, deploy) after customers repeatedly broke production with untested configurations, **reducing production deployment failures by 70%**.
- Engineered Rich Text and Address field types to close competitive gaps in CRM data capture, driving adoption across **15K+ accounts** and cited by sales in enterprise deal wins.
- Diagnosed latency bottlenecks in legacy, high-volume CRM workflows, built REST and bulk Java APIs, and optimized hot paths, **reducing API latency by 35%** while stabilizing the legacy stack.

Projects

SynapseMCP | *Java 21, Spring Boot, Spring AI, PostgreSQL/pgvector, Lucene, Redis, Docker, MCP* GitHub

- Architected a **multi-tenant RAG + MCP platform** where each tenant selects its chat/embedding provider (OpenAI, Anthropic, Ollama, Google GenAI) via REST or a native **15-tool MCP server** (Streamable HTTP), both invoking identical domain services with zero HTTP loopback – proven by a combined REST+MCP integration test suite.
- Engineered hybrid retrieval fusing Lucene BM25 with pgvector cosine ANN via Reciprocal Rank Fusion (k=60) plus an LLM reranking pass to defeat near-duplicate distractors that embeddings alone confuse, lifting Mean Reciprocal Rank from **0.83** (vector-only) to **0.94** (hybrid) to **1.00** (with reranking) on an 18-document golden-query harness.
- Designed a sparse-dimension schema storing six vector widths (384–3072) in one chunks table so tenants on incompatible embedding models share infrastructure with zero runtime DDL, using idempotent HNSW bootstrapping and halfvec to clear pgvector's 2000-dimension index ceiling; validated via EXPLAIN ANALYZE on **313K chunks**.
- Built a five-stage async ingestion pipeline (extract, chunk, embed, persist, index) hardened for partial failure with SHA-256 idempotency, LLM-vision-first extraction with OCR fallback, and hourly self-healing Lucene–PostgreSQL reconciliation.
- Enforced per-tenant Bucket4j rate limiting that degrades to an in-JVM limiter during Redis outages; validated platform behavior with **447+ tests at 94%+ coverage** against real PostgreSQL and Redis instances.

Education

Velammal Institute of Technology August 2017 – August 2021
Bachelor of Engineering (B.E.) in Computer Science *Chennai, India*